

Illustration by **NILANJAN DAS/AI**

THE CHANGE-MAKERS

BHARAT 6G ALLIANCE

- An industry-led, government-backed group, it will focus on how the technology can boost sectors such as health, education, entertainment, agriculture, weather predictions, public services and productivity, among other things

CAPGEMINI

- The tech services firm, in a project with King's College London, has opened a 6G lab near Delhi to prototype, simulate and test solutions, leveraging next-gen connectivity, silicon technologies and advanced AI to address the wireless communication challenges of 6G

DEPARTMENT OF TELECOMMUNICATIONS

- DoT, along with a few IITs and private players, is investing Rs 260 crore to set up two indigenous test beds, a 6G THz (terahertz) one for Orbital Angular Momentum and Multiplexing and another for advanced Optical Communication
- The Centre has given 66 start-ups and MSMEs grants totalling Rs 48 crore to foster digital communication innovations

machine-machine connectivity. By the time India gets to 6G, we will need to support trillions of embeddable devices with trustworthy connections.

The 6G era will have big implications for how governments and industry approach a whole array of themes in public safety and critical asset protection, such as health monitoring, threat detection, facial recognition, traffic analysis, decision-making in areas like law enforcement and social credit systems, besides air quality measurements and weather monitoring. It will also connect the physical world to the human world, thanks to the massive scale deployment of sensors and AI and ML (machine learning) with digital twin models—the virtual replica of an object—and real-time updates. The digital twin models will help analyse what's happening in the physical world, simulate possible outcomes, anticipate needs and then take productive action back into the physical world. Already in use with 5G, digital twin modelling will happen on a much larger scale in 6G.

While the smartphone will remain a key device in the 6G era, new man-machine interfaces will make it more convenient to consume and control information. Gesture and voice control will soon replace touchscreen typing. Devices will be embedded into clothing or even transformed into skin patches. Healthcare will benefit immensely as wearables facilitate 24/7 monitoring of vital parameters.

Additionally, 6G will promote sustainability in myriad ways. By enabling faster and lower cost per bit connectivity, it will support data collection and closed-loop control of appliances. Advanced multi-sensory telepresence and the introduction of multi-modal mixed reality will boost remote collaboration and reduce the need to travel for work. Data processing and data encryption are considered the most critical issues in 6G networks. The advent of 6G will also require a massive expansion of the regulatory structure around wireless spectrum. ■